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Assignment 3

METCS544A2_S2024

Please type or handwrite your answers clearly and concisely. Show all work - credit will not be given for numerical solutions that appear without explanation. **[Total 90 = 3 + 87 pts]**

Grading Rubric

Each question is worth 3 points and will be graded as follows:
3 points: Correct answer with work shown
2 points: Incorrect answer but attempt shows some understanding (work shown)
1 point: Incorrect answer but an attempt was made (work shown)
0 points: Left blank or made little to no effort/work not shown

Reflective Journal [3 pts]
(link to your live google doc)

https://docs.google.com/document/d/1dzAgyy7UhyrmR9CyIVDsXdToMj1JATrVwGZaqPQtogo/edit?usp=sharing

I. Exploring Two Variable Data: Scatterplots and Correlation (8x3 = 24pts)

1) A student wonders if tall women tend to date taller men. She measures herself, her dormitory roommate, and the women in the adjoining rooms; then she measures the next man each woman dates. Here are the data (heights in inches):

Women	66	64	66	65	70	65
Men	72	68	70	68	71	65

a) Is there a clear explanatory variable and response variable in this setting? If so, tell which is which. If not, explain why not.

Answer:

There isn't an explanatory and response variable in this study because its equally likely that the height of a woman influences their choice partner as the height of the man influences their choice of partner.

b) Make a well-labeled scatterplot of these data.

Answer:

c) Based on the scatterplot, describe the pattern, if any, in the relationship between the heights of women and the heights of the men they date.

Answer:

There is a weak positive correlation.

d) Suppose another 70-inch-tall female who dated a 73-in-tall male were added to the data set. How would this influence r?

Answer:

It would increase r.

D

2) From tax records, it is relatively easy to determine the amount of liquor consumed per capita and the number of cigarettes consumed per capita for each of the 10 provinces of Canada. These are plotted on a scatterplot and a high positive correlation is found. Which of the following is correct?

(A) This implies that heavy smoking causes people to drink more.
(B) This implies that heavy drinking causes people to smoke more.
(C) We cannot conclude cause and effect, but this also implies that there is a high positive correlation between cigarette smoking and alcohol consumption for individuals.
(D) This could be an example of a correlation caused by a common cause because both activities are highly correlated with average family income and average income varies widely among the provinces.
(E) We cannot conclude cause and effect, but this also implies that the same individuals both smoke and consume liquor.

C

3) Suppose a study finds that the correlation coefficient relating family income to SAT scores is r = +1. Which of the following are proper conclusions?

I. Poverty causes low SAT scores.
II. Wealth causes high SAT scores.
III. There is a very strong association between family income and SAT scores.

(A) I only
(B) II only
(C) III only
(D) I and II
(E) I, II and III

A

4) An agricultural economist says that the correlation between corn prices and soybean prices is r = 0.7. This means that

(A) when corn prices are above average, soybean prices also tend to be above average.
(B) there is almost no relation between corn prices and soybean prices.
(C) when corn prices are above average, soybean prices tend to be below average.
(D) when soybean prices go up by 1 dollar, corn prices go up by 70 cents.
(E) the economist is confused, because correlation makes no sense in this situation.

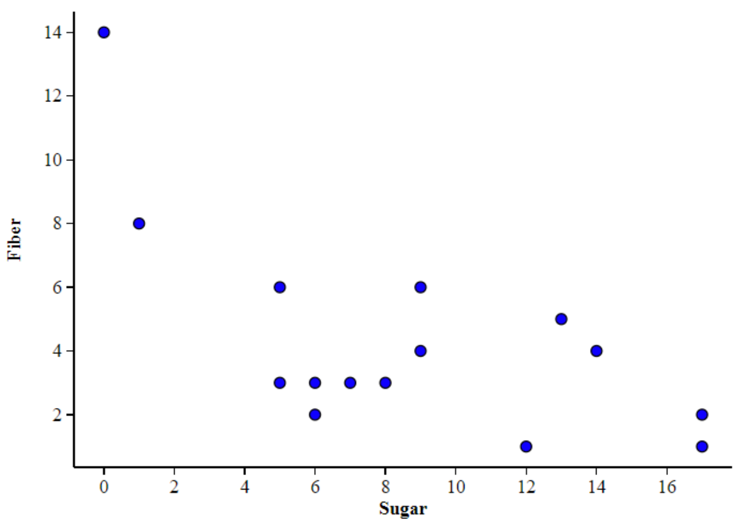
C

5) If data set A of (x, y) data has correlation r = 0.65, and a second data set B has correlation r = –0.65, then

(A) the points in A fall closer to a linear pattern than the points in B.
(B) the points in B fall closer to a linear pattern than the points in A.
(C) A and B are similar in the extent to which they display a linear pattern.
(D) you can't tell which data set displays a stronger linear pattern without seeing the scatterplots.
(E) a mistake has been made—r cannot be negative.

II. Exploring Two Variable Data: Linear Regression (18x3 = 54 pts)

1) Fiber helps regulate the body's use of sugars, helping to keep hunger and blood sugar in check. In children's cereal, which is usually loaded with large amounts of sugar, scientists wanted to investigate if a larger sugar content led to less nutritional value overall, such as lower fiber. A scatter plot of 15 randomly selected children's cereals, along with some selected summary statistics, are given below. Both sugar and fiber are measured in grams per serving.



Sugar	Fiber
$\bar{x} = 8.6$	$\bar{y} = 4.33$
$S_x = 5.18$	$S_y = 3.31$
r = −0.654	

a) Does the scatterplot indicate that it is okay to create a linear model? Explain.

Answer:

The scatter plot doesn't immediately indicate if a linear model is appropriate in this situation. The data looks roughly linear with an outlier at (0,14) but another model might be more applicable, such as exponential decay. Investigating the residual graph would give a better indication if a linear model should be used.

b) Find the slope of the least-squares regression line. Interpret this value in context.

Answer:

slope = $r(S_y/S_x)$ = -0.654(3.31/5.18) = -0.654(0.639) = -0.418
For every additional 1 gram of sugar, there is 0.418 grams less of fiber in children's breakfast cereal.

c) What point must be on the least-squares regression line?

Answer:

The point (8.6 , 4.33) must be on the least-squares regression line.

d) Find the intercept of the least-squares regression line. Interpret this value in context.

Answer:

intercept = y-bar - slope(x-bar) = 4.33 - (-0.418) 8.6 = 4.33 + 3.59 = 7.92
For a typical children's breakfast cereal with zero grams of sugar, the predicted fiber content is 7.92 grams.

e) Write the equation of the linear model.

Answer:

y-hat = -0.418x + 7.92

f) If you pick up a cereal and see that it has 3 grams of sugar, what is the predicted fiber content?

Answer:

y-hat = -0.418 (3) + 7.92 = 6.67

The predicted fiber content is 6.67 grams.

g) What is the residual for the cereal with 9 grams of sugar and 6 grams of fiber?

Answer:

y-hat = -0.418 (9) + 7.92 = 4.158

y-actual = 6

residual = y-actual - y-hat = 6 - 4.158 = 1.842

h) Calculate the value of r^2 and interpret this value in context.

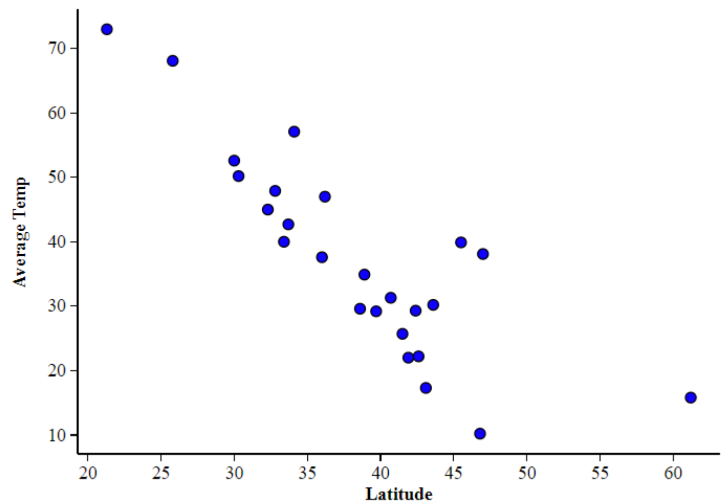
Answer:

$r^2 = (-0.654)^2 = 0.428$

This means that there is a weak correlation between fiber and sugar content in children's cereals.

2) We know that the further you get from the equator, the colder the climate becomes. For the following cities, we have assembled their latitude (angular distance from the equator, measured in degrees) and their average temperature in December of 2021. The data and the scatterplot are given below.

City	Latitude (°N)	Dec Average Temp (°F)
Albany, NY	42.6	22.2
Anchorage, AK	61.2	15.8
Atlanta, GA	33.7	42.7
Austin, TX	30.3	50.2
Bismarck, ND	46.8	10.2
Boise, ID	43.6	30.2
Boston, MA	42.4	29.3
Charleston, SC	32.8	47.9
Chicago, IL	41.9	22
Cleveland, OH	41.5	25.7
Denver, CO	39.7	29.2
Honolulu, HI	21.3	73
Jackson, MS	32.3	45
Knoxville, TN	36	37.6
Las Vegas, NV	36.2	47
Los Angeles, CA	34.1	57.1
Madison, WI	43.1	17.3
Miami, FL	25.8	68.1
Newark, NJ	40.7	31.3
New Orleans, LA	30	52.6
Olympia, WA	47	38.1
Portland, OR	45.5	39.9
Roswell, NM	33.4	40
St. Louis, MO	38.6	29.6
Washington, DC	38.9	34.9



a) Describe the relationship between latitude and average December temperature in the US.

Answer:

There is a negative correlation between latitude and average December temperature in the US.

b) Find the LSRL using Python or R.

Answer:

y-hat = 99.5 - 1.616x

c) Find the following summary statistics using Python or R:

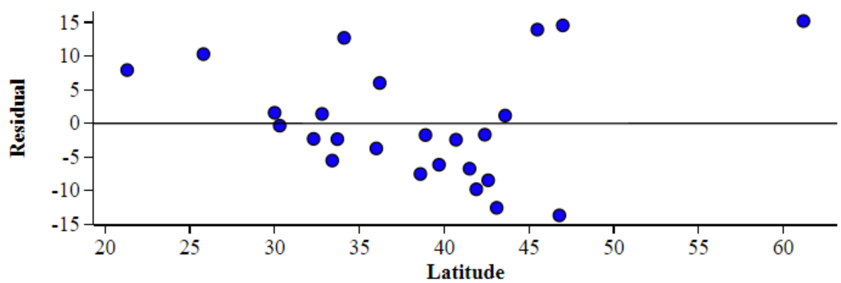
$\bar{x} =$ 38.376 $\bar{y} =$ 37.476 $S_x =$ 8.059 $S_y =$ 15.544 r = -0.838

d) Using the summary statistics from (c), show that the LSRL you find matches what you found in (b).

Answer:

$b_1 = r(S_y/S_x) = -0.838(15.544/8.059) = -0.838(1.929) = -1.616$
$b_0 = \text{y-bar} - b_1 \text{ x-bar} = 37.476 - (-1.616)\ 38.376 = 37.476 + 62.024 = 99.5$

e) Below is the residual plot for latitude and average December temperature.



Given the residual plot above, is a linear regression model appropriate for this data set? Explain.

Answer:

A linear regression model is appropriate because the residuals are mostly randomly distributed. There is no clear pattern which suggests a different model would be better.

f) Kansas City, MO is located at 39.1°N and had an average December 2021 temperature of 42°F. What is the residual for this point?

Answer:

y-hat = 99.5 - 1.616 (39.1) = 99.5 - 63.1856 = 36.314
y-actual = 42
residual = 42 - 36.314 = 5.686

III. Exploring Two Variable Data: Influential Points (3x3 = 9 pts)

1) A sample of men agreed to participate in a study to determine the relationship between several variables including height, weight, waist size, and percent body fat. A scatterplot with percent body fat on the y-axis and waist size (in inches) on the horizontal axis revealed a positive linear association between these variables. Computer output for the regression analysis is given below:

Dependent variable is: %BF
R-squared = 67.8%
S = 4.713 with 250-2 = 248 degrees of freedom

Variable	Coefficient	se of coeff	t-ratio	prob
Constant	-42.734	2.717	-15.7	<.0001
Waist	1.70	0.0743	22.9	<.0001

(a) Write the equation of the regression line:

Answer:

y-hat = 1.70x - 42.734

(b) Explain/interpret the information provided by R-squared in the context of this problem. Be specific.

Answer:

The R-squared value suggests there is a weak correlation between waist size and body fat. The positive slope indicates a positive relationship between the variables. Overall the data suggests that men with larger waist sizes tend to have greater body fat.

(c) One of the men who participated in the study had a waist size 35 inches and 10% body fat. Calculate the residual associated with the point for this individual.

Answer:

y-hat = 1.70 (35) - 42.734 = 59.5 - 42.734 = 16.766
y-actual = 10
residual = 10 - 16.766 = -6.766

THE END